

March 25, 2026

Journal of Machine Learning Research

Dear Editors,

I am submitting “Cognitive Dynamics of an Epistemically Constrained Language Model Agent” for consideration at JMLR.

The paper presents empirical evidence that a language model agent equipped with persistent metacognitive infrastructure produces cognitive state trajectories with learnable temporal dynamics. Over five months of continuous operation, the system logged 68,110 evaluation ticks aggregated to 2,992 turn-level state snapshots. A neural predictor trained on these trajectories beats the persistence baseline by 41.7%. The strongest signal appears in rare-event features with near-zero autocorrelation, indicating genuine transition dynamics rather than momentum. Adding user input features provides no predictive improvement over internal state alone, suggesting autonomous dynamics driven by architectural constraint. Intra-turn analysis reveals a bimodal adaptive deliberation mechanism with two distinct processing regimes separated by a sharp phase boundary.

1. **Prior publications.** This work has not been previously published in any form. There are no conference papers, workshop papers, or preprints that overlap with this submission.
2. **Co-author consent.** I am the sole author and consent to review by JMLR.
3. **Conflicts of interest.** I have no conflicts of interest with any JMLR action editors or editorial board members. I have no institutional affiliation and have not collaborated with any members of the JMLR editorial board. No third-party funding was received for this work. No financial relationships exist with entities that could be perceived to influence the submitted work. The manuscript acknowledges the use of Claude (Anthropic) for data analysis pipeline development and manuscript preparation assistance; this was through standard consumer product access, not a sponsored or funded relationship.
4. **Suggested Action Editors:**
 - (a) Maxim Raginsky (UIUC) — expertise in dynamical systems, statistical learning, and applied probability aligns with the paper’s framing of agent cognitive state as a dynamical system with empirically measured trajectories and phase boundaries.
 - (b) Honglak Lee (Google / U. Michigan) — expertise in representation learning and unsupervised learning connects to the self-supervised state prediction methodology and proposed extensions to learned cognitive state representations.
 - (c) George Konidaris (Duke) — expertise in reinforcement learning, AI, and robotics aligns with the paper’s connections to world model frameworks and embodied agent architectures.
 - (d) Csaba Szepesvari (U. Alberta) — expertise in reinforcement learning and sequential decision making connects to the temporal state prediction methodology.
 - (e) Kai-Wei Chang (UCLA) — expertise in large language models and trustworthy NLP aligns with the paper’s focus on LLM agent monitoring and behavioral characterization.

I have no conflicts of interest with any of the suggested action editors.

5. **Suggested Reviewers:**

- (a) Nicolas Tacheny — author of work on geometric dynamics of agentic loops in LLMs (arXiv:2512.10350), directly adjacent to the dynamical systems framing used in this paper.
- (b) Yukun Zhang — co-author of the DMET framework for latent representational dynamics in LLMs (arXiv:2505.20340), relevant expertise in modeling LLM behavior as dynamical systems.
- (c) David Abel (JMLR editorial board reviewer) — listed expertise in reinforcement learning, philosophy, planning, and abstraction.
- (d) Abhishek Rath — author of the agent drift framework (arXiv:2601.04170), directly complementary work on temporal behavioral characterization of LLM agent systems.
- (e) Chris Amato (JMLR editorial board reviewer) — listed expertise in multi-agent reinforcement learning and multi-robot systems.

I have no conflicts of interest with any of the suggested reviewers.

6. **Keywords:** cognitive dynamics, agent architecture, temporal state prediction, metacognitive monitoring, adaptive deliberation, world models, epistemic constraint

Sincerely,

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